Unit 15 Overview: Probability

Unit Narrative

In this unit, students will explore probability in a variety of ways. In Section 1, students will explore how to find the sample space of a simple experiment. Students learn that in order to determine the sample space of a simple experiment they must find all the outcomes that are possible and write them as a set or list. This idea will be important in Section 2 when students learn how to find the probability of an event in a simple experiment.

Section 1 lays the foundation for understanding how to determine the sample space of a simple experiment and probability. In Section 2, students apply what they have learned to find the probability of simple experiments. Students start with an introduction to probability and the likelihood of an event occurring. Students learn that probability is on a scale from 0 to 1 and can be expressed as a decimal, fraction, or percentage. Students will also learn how to describe an event by its likelihood of occurring. Once students have an understanding of finding simple probability of an event, they learn the difference between theoretical and experimental probability. This is done through performing a series of experiments and simulations.

In Grade 7 Accelerated Math, students will expand on these ideas to determine the sample space and probability of repeated and two-step experiments.

Unit At-A-Glance

Section 1: Exploring Sample Space in Probability	Benchmarks Addressed	Lesson 1	Exploring Sample Space
	MA.7.DP.2.1	Lesson 2	Determining Sample Space
Section 2: Finding Probability	Benchmarks Addressed	Lesson 3	Probability
	MA.7.DP.2.2 **MA.7.DP.2.3** **MA.7.DP.2.4**	Lesson 4	Comparing Probabilities
		Lesson 5	Experimental and Theoretical Probability
		Lesson 6	Simulating Small and Large Numbers of Trials
		Lesson 7	Experimental and Theoretical Probabilities from Simulations

Differentiation

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Engagement *The "WHY" of learning*	• To recruit student interest in Lesson 7, have students research the statistics for a sports player of their choosing. Students can then work together to come up with a simple experiment that could be used to simulate the probability of the event researched.
Representation *The "WHAT" of learning*	• As gaps in mastery are revealed, consider using the On-Ramp tool to address specific gaps and determine the best way to scaffold throughout the remainder of this unit. • In Lesson 6, for an alternative approach to 15.6.4 Try it, consider creating a Card Sort out of the descriptions and simulations. Students can then physically match the descriptions to the simulations with their partner.
Action & Expression *The "HOW" of learning*	• In Lesson 4, consider a kinesthetic approach to 15.4.3 question 2. Read the problem aloud and have students move to one side of the room or the other depending on which statement they think is more likely. If they think they are equally likely, they can stand in the middle of the room. • In Lesson 2, 15.2.2 Exploration, for a kinesthetic and visual approach to this activity, provide partner pairs with a coin and a number cube. This allows students the opportunity to visualize the sample space by completing trials of the experiment.

ELL Information

Close Reading Strategy
Close Reading is an effective decoding strategy for building literacy and supporting students early in their language acquisition stages. Similar to the Three Reads Language Routine, students intentionally target specific pieces of information in order to scaffold the decoding process and reduce anxiety about word problems.
Consider opportunities to incorporate Close Reading:

- Lesson 1: 15.1.1 Warm-Up
- Lesson 2: 15.2.2 Exploration
- Lesson 4: 15.4.3 Guided Practice

All glossary terms are available in multiple languages, including English, Spanish, Haitian Creole, and American Sign Language, as support for ELL students. Consider pairing the following videos with strategies from “Additional Differentiated Support.”

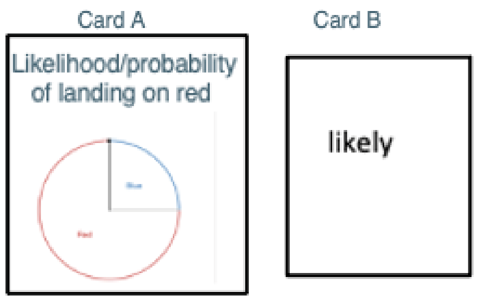
- Event
- Experimental probability
- Sample space
- Simulation
- Theoretical probability

Additional Differentiated Support

Scaffolding Support

Card Sort

Example Card Set:



Create sets of matching cards with one card having an image of a chance experiment, and the other card having the likelihood or probability of the event.
You could also add the probability of the event to Card B as well as the likelihood.
Students can use these cards in a variety of ways:

- Play memory by flipping the cards over and trying to get a match.
- Playing “Go Fish” with a partner to find the matching points with their distance on the coordinate plane.
- Students can time themselves and see how fast they can find the matching cards.

On-Ramp

On-Ramp to 6th Grade Math provides video and practice tools for students to scaffold the following skills:

- Recognize and Generate Equivalent Fractions Using Models
- Compare and Order Decimals with Models

Section Coherence

Section 1: Exploring Sample Space in Probability (Lessons 1 – 2)

<table>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="2">MA.7.DP.2.1 Determine the sample space for a simple experiment.
Clarification 1: Simple experiments include tossing a fair coin, rolling a fair die, picking a card randomly from a deck, picking marbles randomly from a bag and spinning a fair spinner.</td><td>Lesson 1</td><td>I can determine all possible outcomes of a given scenario.
I can determine whether or not a simple experiment is fair.</td></tr>
<tr><td>Lesson 2</td><td>I can determine the sample space for a simple experiment.</td></tr>
</table>

Section 2: Finding Probability (Lessons 3-7)

<table>
<tr><th>Benchmarks Addressed</th><th colspan="2">Learning Targets</th></tr>
<tr><td rowspan="5">MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.
Clarification 1: Instruction includes representing probability as a fraction, percentage or decimal between 0 and 1 with probabilities close to 1 corresponding to highly likely events and probabilities close to 0 corresponding to highly unlikely events.
Clarification 2: Instruction includes $P(event)$ notation.
Clarification 3: Instruction includes representing probability as a fraction, percentage or decimal.
MA.7.DP.2.3 Find the theoretical probability of an event related to a simple experiment.
Clarification 1: Instruction includes representing probability as a fraction, percentage or decimal.
Clarification 2: Simple experiments include tossing a fair coin, rolling a fair die, picking a card randomly from a deck, picking marbles randomly from a bag and spinning a fair spinner.
MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.
Clarification 1: Instruction includes representing probability as a fraction, percentage or decimal.
Clarification 2: Instruction includes recognizing that experimental probabilities may differ from theoretical probabilities due to random variation. As the number of repetitions increases experimental probabilities will typically better approximate the theoretical probabilities.
Clarification 3: Experiments include tossing a fair coin, rolling a fair die, picking a card randomly from a deck, picking marbles randomly from a bag and spinning a fair spinner.</td><td>Lesson 3</td><td>I can express the probability of a chance event using a fraction, percentage, or decimal.
I can compare probabilities of chance events.</td></tr>
<tr><td>Lesson 4</td><td>I can interpret the likelihood of a chance event occurring using probability.
I can compare probabilities of chance events.</td></tr>
<tr><td>Lesson 5</td><td>I can perform a simulation to determine an experimental probability.
I can find the theoretical probability of an event.
I can compare the experimental probability from a simulation to the theoretical probability of a simple experiment.</td></tr>
<tr><td>Lesson 6</td><td>I can compare the experimental probability from a simulation to the theoretical probability of a simple experiment.
I can use results from computer-simulated event to informally define the Law of Large Numbers.</td></tr>
<tr><td>Lesson 7</td><td>I can determine the probability of an event from a simulation and compare it to the theoretical probability of the event.</td></tr>
</table>

Vertical Alignment

	Grade 6 Accelerated Unit 15 Probability	Grade 7 Accelerated Unit 17 Probability
Past Course(s)	**Course Level**	**Future Course(s)**
	MA.7.DP.2.1 Determine the sample space for a simple experiment. **MA.7.DP.2.2** Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events. **MA.7.DP.2.3** Find the theoretical probability of an event related to a simple experiment. **MA.7.DP.2.4** Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.	**MA.8.NSO.1.1** Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line. **MA.8.DP.2.1** Determine the sample space for a repeated experiment. **MA.8.DP.2.2** Find the theoretical probability of an event related to a repeated experiment. **MA.8.DP.2.3** Solve real-world problems involving probabilities related to single or repeated experiments, including making predictions based on theoretical probability.

15.1 Exploring Sample Space

Lesson Introduction

Benchmark	MA.7.DP.2.1 Determine the sample space for a simple experiment.	Learning Targets	• I can determine all possible outcomes of a given scenario. • I can determine whether or not a simple experiment is fair.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.8.DP.2.1 Determine the sample space for a repeated experiment.

Essential Questions	• How does the number of possible outcomes for an experiment relate to the likelihood of a particular event? • How can we apply probability to inform a prediction?
Lesson Narrative	This lesson is the first within the unit, which introduces probability. The lesson engages students in thinking about sample space by using contexts they are already familiar with – games and fairness. While key vocabulary related to MA.7.DP.2.1 is introduced, this lesson is also designed to hook students by using their prior knowledge and experience relating to probability. In Lesson 2, students will delve more explicitly into determining sample spaces for simple experiments.

Component Summary	15.1.1 Warm-Up (~5 min)	Estimation activity using benchmark percentage of 50% (SE pg. 401)	15.1.3 Exploration (5 min)	Students discover how “fairness” relates to probability (SE pg. 404)
	15.1.2 Guided Instruction (10 min)	Introduction of key probability terms (SE pg. 402)	15.1.4 Activity (20-25 min)	Students compare sample spaces and fairness of carnival games in a gallery walk activity (SE pg. 404)
	15.1.5 Wrap-Up (~5 min)	Students reflect on their understanding of the lesson (SE pg. 405)		

	Glossary	Materials	Additional Preparation
Resources	Key Terms: • Outcome • Sample space • Event Additional Terms: N/A	• Gallery Walk images • (Optional) Example game pieces to model gallery walk games	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>Students may believe that all outcomes are equally likely in all scenarios.
Students may believe that future outcomes in these experiments depend on previous outcomes (all examples in this lesson are of independent probability).
Students may confuse the terms “outcome” and “event.”</td><td>In 15.1.3 Exploration, consider asking students to make the explicit connection between fractional sizes of the spinner sections and the “fairness” of the spinners.
Model choosing a card from Doug’s chore list to demonstrate the difference between “outcome” and “event.” Using repetition can help students solidify the vocabulary.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Number Talk
The 15.1.1 Warm-Up contains a question that asks students to estimate a percentage given a fraction and explain their reasoning. Allow each student time to independently come up with a strategy for estimating over or under 50%. Then, select students to share out different strategies they used in solving the problem. Consider adjusting the time for the Warm-Up to 10 minutes if utilizing this strategy.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
The 15.1.4 Activity is a collaboration opportunity for students. Students get to work together to review carnival game rules and determine whether or not those games should be considered “fair.” Student pairs also have the opportunity to discuss and reflect on their thinking with other student groups at the conclusion of the activity.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The carnival games presented in 15.1.4 require students to read the rules of a game and determine the possible outcomes and “fairness” of that game. Auditory learners would benefit from hearing the verbal description of the activities and both kinesthetic and visual learners would benefit from watching a demonstration of the game.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

15.1.1 Warm-Up: Carolyn Beatrice Parker

Component Overview: *Students practice using mental math strategies to assess a percentage value relative to the benchmark percentage of 50%.*

15.1.1 Warm-Up: Carolyn Beatrice Parker

Carolyn Beatrice Parker, born in Gainesville, FL in 1917, was the first African American woman known to receive a graduate degree in physics and subsequently work on the top-secret government project known as the Manhattan Project. Before earning her master's degree in physics, she earned a master's degree in mathematics and was also a teacher in Florida.

In 2020, an elementary school in Gainesville, FL was re-named in her honor. At this time, 11 of the 21 elementary schools in the Alachua County Public School system were named after people.

1. Without performing any calculations, select the statement that best describes the percentage of the elementary schools in Alachua County that were named after people as of 2020.

   - significantly less than 50%
   - slightly less than 50%
   - exactly 50%
   - slightly more than 50%
   - significantly more than 50%

2. Explain your reasoning.
   Answers will vary. The statement that best describes the percentage is slightly more than 50% because 21 is close to 20 and 10 is 50% of 20, therefore 11 is slightly more than 50% of 21.

Number Talk
Consider also allowing students to discuss their reasoning with a partner first before bringing the class back for a whole-group discussion. This practice can help students clarify their mathematical thinking.

- *Do you know any schools in your area named after someone famous?*

15.1.2 Guided Instruction: Determining Outcomes

Component Overview: *Students will begin practicing creating sample spaces to display the total possible outcomes for an experiment. They will identify the number of outcomes in a particular event. Activities are designed to build familiarity with probability vocabulary.*

15.1.2 Guided Instruction: Determining Outcomes

An ***outcome*** is a possible result of an experiment.

1. A spinner divided into four equal areas is shown.

   a. How many outcomes are there from spinning the spinner?
   4

In a probability model for a random process, the ***sample space*** is a list of the individual outcomes that are being considered.

   b. With your partner, discuss the sample space for the spinner. Write a list of the possible outcomes.
   B, R, G, Y

An ***event*** is a set of possible outcomes resulting from an experiment. In general, an event is any subset of a sample space.

   c. How many outcomes are included in the event that the spinner lands on blue?
   One

   d. Primary colors include yellow, blue, and red. How many outcomes are included in the event that the spinner lands on a primary color?
   Three

- *How do you find the total number of outcomes?*
- *Does one outcome seem more likely than another?*

Students are becoming familiar with probability vocabulary for the first time. To have additional practice understanding the concept of sample space, consider prompting students to discuss the sample spaces for different experiments: outcomes of a Miami Heat game, shoes to wear to school, rolling a die, flipping a coin, etc.

2. Doug's parents created a set of cards with chores on them. Each Saturday, Doug chooses a card to determine his chore.

Mow the yard	Dust and vacuum the house	Clean the bathrooms	Clean your bedroom
Clean your bedroom	Mow the yard	Weed the garden	Mow the yard
Clean the bathrooms	Parents' choice	Dust and vacuum the house	Free day

a. What are the possible outcomes for the chore cards?
Mow the yard, Dust and vacuum the house, Clean the bathrooms, Clean your bedroom, Weed the garden, Parents' choice, Free day

b. Which outcomes have more than one card?
Mow the yard, Dust and vacuum the house, Clean the bathrooms, Clean your bedroom

c. On a particularly hot Saturday, Doug is hoping he doesn't select an outdoor chore. Which outcomes could be included in the event that he chooses an outdoor chore?
Mow the yard, Weed the garden

Students should be able to complete question 2 independently. Consider having them check in with a partner to compare responses.

- *Which outcome has the most cards?*
- *Which outcome(s) have the least cards?*

For English Language Learners, determining what constitutes an "outdoor" vs. "indoor" chore may be challenging to do independently. Consider providing a visual for each card as support for this activity.

Consider asking the question, "if Doug chooses a card at random, which chore do you think he is most likely to get?" Students can begin to make connections between the number of outcomes and the probability of an event.

15.1.2 Guided Instruction: Determining Outcomes (continued)

d. How many outcomes could be included in the event that Doug chooses an indoor chore?
6 - Dust and vacuum the house (2 cards), Clean the Bathrooms (2 cards), and Clean your bedroom (2 cards)

3. Use the word bank to complete the summary of key concepts below.

event	outcomes	sample space	set

To determine the ____*sample space*____ of a simple experiment, find all the ____*outcomes*____ that are possible and write them as a ____*set*____.

An ____*event*____ is a subset of the sample space, such as rolling an even number on a 6-sided die.

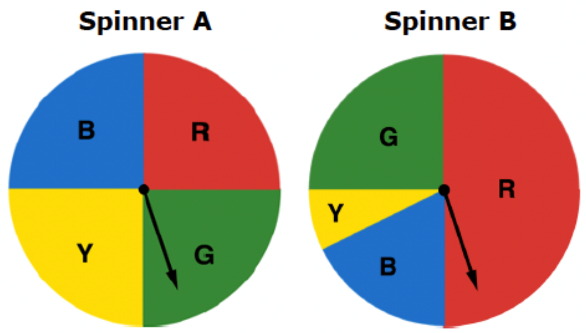
Clarify in question 3 that "set" and "list" can be used interchangeably in this context.

15.1.3 Exploration: Digging Deeper Into Outcomes

Component Overview: *Students will begin exploring the concept of "fairness" by comparing two spinners with the same total number of outcomes, but different likelihoods associated with each outcome.*

15.1.3 Exploration: Digging Deeper into Outcomes

Two spinners are shown.



Allow students to discuss which spinner is more favorable for each outcome. For example, pose the question, "If you wanted a spinner to land on red, which spinner would you prefer to use?" Encourage students to use mathematical support for their responses.

1. Work with your partner to compare and contrast the features of the two spinners.

Similarities	Differences
size of the spinner amount of sections name and colors of the sections	size of the sections except for G

- *What does it mean for the spinner to be "fair?"*
- *Can you give an example of a fair game?*
- *An example of a game that is not fair?*

2. One of the spinners is considered a fair spinner. Complete the sentence by selecting the fair spinner and writing an explanation.

○ Spinner A
○ Spinner B

is the fair spinner because all of the sections have the same chance to be selected.

15.1.4 Activity: Gallery Walk

***Component Overview:** Students will analyze four different carnival games to determine the possible outcomes of each game and whether or not the game would be considered "fair." Post the activities around the room and allow students to circulate among them. Alternatively, partner pairs can receive copies of the carnival games to complete the activity.*

15.1.4 Activity: Gallery Walk

1. Work with your partner to determine the sample space for each game and whether the game is fair. Fill in the chart with your answers.

Game Name	Sample Space	Is it fair?
Skee-Ball	*0, 10, 20, 30, 40, 50, 100*	*No*
Card Draw	*Clubs- Ace,2,3,4,5,6,7,8,9,10, Jack, Queen, King* *Spades- Ace,2,3,4,5,6,7,8,9, 10, Jack, Queen, King* *Hearts- Ace,2,3,4,5,6,7,8,9, 10, Jack, Queen, King* *Diamonds- Ace,2,3,4,5,6,7,8, 9, 10, Jack, Queen, King*	*Yes*
Prize Wheel	*orange, blue, red, yellow, white*	*No*
Rubber Ducks	*1,2,3,4,5,6,7,8,9,10,11,12,13, 14,15,16,17,18,19,20,21,22, 23,24,25*	*Yes*

2. Which game has the largest sample space?
Card Draw

3. Discuss with your partner which game's sample space was the most difficult to determine.
Answers will vary.

This activity requires students to read the rules for a game to determine whether or not the game is fair. For students who struggle with reading comprehension, it may be helpful to provide a demonstration of how each game is played. Consider splitting the class into four groups to simulate a few rounds of each game.

Consider asking students to make rule proposals for the games they deemed unfair. How can the rules be modified to make these games fair?

Why do you think a carnival would choose to offer both fair and unfair games?

15.1.4 Activity: Gallery Walk (continued)

4. Carnivals include many games similar to these. What do you notice about the fairness of such games?
Answers will vary. I notice that some games are noticeably fair or unfair, while other games are unnoticeably unfair.

5. Find another pair of partners to form a group of four. Compare your responses on whether each game is fair and discuss your reasoning. Record one similarity and one difference you notice in your reasoning.
Answers will vary.

Similarity	Difference
We all thought that the Card Draw was fair because all of the suits had an equal chance of being drawn.	*We thought that the Prize Wheel was not fair because all of the colors are not the same size, and our partners thought that the Prize Wheel was fair because all of the colors are the same size.*

6. Future lessons in this unit will explore probabilities with fair coins, fair dice, and fair spinners. Describe what you think it means for a coin, a die, or a spinner to be fair.
Answers will vary. Fair means that all of the outcomes have the same chance to be selected.

Provide an opportunity for meaningful reflection with questions 3 and 4 on "fairness." This response could be used for you to evaluate student understanding from the lesson.

15.1.5 Wrap-Up: Reflection

Component Overview: *Students complete a self-reflection on their level of understanding of the content of this lesson.*

15.1.5 Wrap Up: Reflection

Complete each statement based on your understanding at this point.
Answers will vary.

1. I am still unsure about ____________________________

   ____

   ____.

2. One question I have that will help me better understand

   is ____

   ____

   ____?

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

15.2 Determining Sample Space

Lesson Introduction

Benchmark	MA.7.DP.2.1 Determine the sample space for a simple experiment.	Learning Target	• I can determine the sample space for a simple experiment.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.8.DP.2.1 Determine the sample space for a repeated experiment.

Essential Questions	• How do the number of possible outcomes for an experiment relate to the likelihood of a particular event? • How can we apply probability to inform a prediction?
Lesson Narrative	In this lesson, students begin to assign probabilities to chance events and identify the sample spaces for simple experiments. They build the understanding that the greater the probability, the more likely an event is to occur. Furthermore, students determine whether outcomes in a simple experiment are "equally likely," building on the concept of fairness explored in the previous lesson.

Component Summary	15.2.1 Warm-Up (~5 min)	Students choose between two games to play (SE pg. 406)	15.2.3 Bring It Together (5-10 min)	Students reflect on exploration and apply new vocabulary (SE pg. 409)
	15.2.2 Exploration (15-20 min)	Deeper exploration into games from the Warm-Up and discovery of sample space (SE pg. 407)	15.2.4 Guided Practice (10 min)	Practice finding the sample space in a simple experiment (SE pg. 409)
	15.2.5 Wrap-Up (~5 min)	Formative assessment on finding the sample space in a simple experiment (SE pg. 410)		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Simple experiment • Probability <u>Additional Terms:</u> • Sample space	• Letter cards for 15.2.2 Exploration • (Optional) Probability manipulatives- coins, dice, spinners	N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may think that the phrase “equally likely” means that there is a 50% chance of an outcome happening.
Students may think that if a specific outcome has not occurred when they attempted an experiment, it means that outcome is impossible.</td><td>In 15.2.2 Exploration, model understanding of “equally likely” outcomes by changing the questions to ask about different outcomes in the sample space. For example, ask about the probability of rolling a 1 on a die, then a 4, then a 5, etc.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
15.2.3 Bring It Together is a summary activity of the partner Exploration. In the activity, students are asked to consider the likelihood of different outcomes from the paper activity. Allow each student to make a conjecture about the answer to the questions and share with a partner to explain their reasoning. Listen as students are sharing to support in summarizing the key takeaways of the exploration.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
15.2.2 Exploration is a collaboration opportunity for students. Students get to work together to compare the likelihood of winning between two games. They will analyze the mathematical thinking of others and justify their reasoning, gaining important practice in this Mathematical Thinking and Reasoning Standard.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">The simple experiments explored in 15.2.1 Warm-Up and 15.2.2 Exploration contain examples of common probability games. Consider preparing sets of manipulatives (dice, coins, and spinners). The use of manipulatives offers kinesthetic and visual entry points for learning about outcomes and sample spaces.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

15.2.1 Warm-Up: Which Game Would You Choose?

Component Overview: *Students analyze two games to determine which game they would rather play. In this Warm-Up, students apply their innate understanding of probability.*

15.2.1 Warm-Up: Which Game Would You Choose?

1. Two games are described.

   Game 1: A player flips a coin and wins if it lands on heads.

   Game 2: A player rolls a standard number cube and wins if it lands on a number that is divisible by 3.

   a. Which game would you rather play?
   Answers will vary. I would rather play Game 1. I would rather play Game 2.

   b. Explain your reasoning.
   I would rather play Game 1 because my chances of winning are equal to my chances of losing. I would rather play Game 2 because there is more than one chance of winning.

Use this Warm-Up as a way to gauge students' understanding about probability from their experience playing games. Students do not need to use formalized language of probability to answer this question. Allow students to share their responses, but do not provide a "correct" answer.

15.2.2 Exploration: Simple Experiments

Component Overview: *Working with partners, students make connections between a simple game's sample space, possible outcomes, and the probability of winning. Students review the example games from the Warm-Up and discover how to prove mathematically which game has a greater likelihood of a player winning. In the second half of the Exploration, students are introduced to the idea that not all sample spaces are obvious before actually performing the experiment.*

15.2.2 Exploration: Simple Experiments

Work with your partner to complete the following.

1. Discuss the sample space for both games from the Warm-Up with your partner.

   a. Write the sample space for each in the table.

Random Process	Sample Space
Flipping a Coin 	*heads, tails*
Rolling a Standard Number Cube 	*1, 2, 3, 4, 5, 6*

   b. What are the possible outcomes in the event of landing on heads in Game 1 from the Warm-Up?
   Heads

   c. What fraction of the possible outcomes from Game 1 results in a win?
   $\frac{1}{2}$

   d. What are the possible outcomes in the event of rolling a number divisible by 3 in Game 2 from the Warm-Up?
   three, six

   e. What fraction of the possible outcomes from Game 2 results in a win?
   $\frac{1}{3}$

For a kinesthetic and visual approach to this activity, provide partner pairs with a coin and a number cube. This allows students to visualize the sample space by completing trials of the experiment.

How did you figure out the sample space for each random process?

15.2.2 Exploration: Simple Experiments (continued)

f. Based on your answers to Parts B-E, complete the statement.
A player has a greater likelihood of winning

- Game 1
- Game 2

because $\frac{1}{2} > \frac{1}{3}$.

g. How could Game 2 be modified so the likelihood of winning either game is the same?
Answers will vary. Game 2 can be modified so that a player can win if the standard number cube lands on an even number.

2. With your partner, determine who is partner A and who is partner B. For this activity, you and your partner will pull slips of paper from a bag.

a. Make a guess about what is on the slips of paper.
Answers will vary. I guess numbers are on the slips of paper.

b. Without looking in the bag, partner A should take out one of the papers and show it to partner B. Both partners record what was printed on the paper in the first row of the table.
Answers will vary.

Who?	What is printed on the paper?
Partner A, Choice 1	
Partner B, Choice 1	
Partner A, Choice 2	
Partner B, Choice 2	
Partner A, Choice 3	
Partner B, Choice 3	

MA.K12.MTR.4.1: Discussion and Reflection
Consider having a whole class discussion as to why the larger fraction indicates a greater likelihood of winning. This could illuminate potential misconceptions on how students understand probability.
Ideally, students will understand that the closer the fraction is to 1, which means that winning is certain, the more likely that game is to produce a win.

For an additional challenge, ask students if Game 1 could be modified to match the likelihood of Game 2.

Provide each partner pair with a paper bag containing the cut-up letter strips from the Blackline Master.

Demonstrate the steps for the activity by acting out an example round. This can help students process how the activity works before attempting it.

15.2.2 Exploration: Simple Experiments (continued)

c. Partner A should put the paper back into the bag. Then, partner B should take out one of the papers and show it to partner A for both to record in the table.

d. After both partners have made their first selection and put it back in the bag, make a revised guess of the sample space.
Answers will vary. I guess letters are on the slips of paper.

e. Continue taking turns and recording the results in the table until each partner has taken three turns.

f. Make a revised guess of the sample space.
Answers will vary. I guess letters are on the slips of paper.

g. Rate your confidence in your last guess using the emoji scale.
Answers will vary.

h. Discuss with your partner what you could do to make a better guess of the sample space. Write at least two strategies you discussed.
Answers will vary. I can take all of the papers out of the bag. I can also ask another classmate.

i. Look at all the papers in the bag. Write the sample space.
A, B, C, D, E, F, G, H, I, J, K, L, M, N, O

- *What would be helpful to make a more accurate guess of the sample space?*
- *Can you tell if all outcomes are equally likely?*
- *How did you refine your sample space guesses for each round?*
- *Were any of your guesses correct for the sample space?*
- *What information would have helped you make a better guess?*

If time allows, consider facilitating a longer discussion about outcomes and sample spaces. Ask students to think about having a new bag of papers and taking out a strip 100 times. If they never pulled out the letter “Q,” would this necessarily mean that “Q” was not a part of the sample space? What could they conclude?

15.2.3 Bring It Together: Defining Simple Experiments

Component Overview: *Students are introduced to formal vocabulary "simple experiment" and "probability." Students confirm the games they explored previously are simple experiments and compare the likelihood of different outcomes in one of the games.*

15.2.3 Bring It Together: Defining Simple Experiments

The games from the exploration are all considered **simple experiments**.

A ***simple experiment*** is an action that has a definite outcome that cannot be predicted with certainty.

In a simple experiment, each outcome in the sample space is equally likely to happen; however, events, which are subsets of the sample space, may have different likelihoods of happening. This is called probability.

The ***probability*** of an event is a measure of the likelihood the event will occur.

1. Consider the simple experiment involving pulling papers from the bag in the Exploration.

   a. Explain whether all of the possible outcomes in the sample space were equally likely or not.
   Answers will vary. All of the possible outcomes were equally likely because there is only one of each letter.

   b. Explain which of the following events would have been more likely to happen in that experiment – pulling out a paper with a vowel on it or pulling out a paper with a consonant on it.
   Answers will vary. Pulling out a paper with a consonant on it would have been more likely because there are more consonants than there are vowels.

- *What does it mean to say that "each outcome in the sample space is equally likely to happen?"*
- *Can you give an example of an experiment where each outcome is not equally likely to happen?*

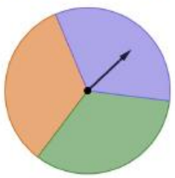
Think-Pair-Share
For question 1b, consider using a think-pair-share activity. Allow students independent thinking time to consider the two different outcomes and write a response. Then, they can share their thinking with a partner and revise their written responses if needed. Students will then share out as a whole group.

15.2.4 Guided Practice: Determining Sample Space of Simple Experiments

Component Overview: *Students practice finding the sample space of simple experiments and applying their knowledge of probability to create a fair game.*

15.2.4 Guided Practice: Determining Sample Space of Simple Experiments

1. For each simple experiment in the table, determine the sample space.

Simple Experiment	Sample Space
Spinning the spinner 	*orange, purple, green*
Rolling a fair 12-sided die 	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12

Students can complete the first task independently. Consider starting the second task independently and allowing students to check their work with a partner before reviewing as a whole group.

- *Are the outcomes in these experiments "equally likely?"*

15.2.4 Guided Practice: Determining Sample Space of Simple Experiments (continued)

2. Complete the following.

   a. Given the sample space of {Reese's, Snickers, Twix, Kit Kat}, label the game wheel so that there is a fair chance of winning each candy bar as a prize.

   b. Explain how you determined the labels for the game board.
   Answers will vary. I divided the number of sections on the game wheel, 12, by the number of outcomes in the sample spaces, 4. I labeled the wheel with each outcome 3 times to be sure that all outcomes have equal chances of winning.

Consider scaffolding 2a to assist students who may have struggled in 15.2.2 Exploration and 15.2.3 Bring It Together.
Suggested questions:
- *How many outcomes are on the spinner?*
- *How many outcomes are in the sample space?*
- *What does it mean for there to be a fair chance of winning each candy bar?*

Provide an opportunity for meaningful reflection with question 2b. Allowing students to hear one another's thought processes can solidify their thinking on what it means for outcomes to be "equally likely." This response could be coupled with the Wrap-Up for you to evaluate student progress from the lesson.

15.2.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students complete a formative assessment on identifying the sample space of a simple experiment.*

15.2.5 Wrap-Up: Ticket Out the Door

1. For each simple experiment in the table, determine the sample space.

Simple Experiment	Sample Space
Selecting a day of the week from a deck of cards with each day written on one card	*Sunday, Monday, Tuesday, Wednesday, Thursday, Friday, Saturday*
Flipping an integer chip 	*yellow, red*

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

Consider asking students to compare the likelihood of the outcomes in the given simple experiments. Could students modify the experiments so that the outcomes are not "equally likely?"

15.3 Probability

Lesson Introduction

Benchmark	MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.	Learning Targets	• I can express the probability of a chance event using a fraction, percentage, or decimal. • I can compare probabilities of chance events.

	Building On	Working Toward
Benchmark Coherence	N/A	N/A

Essential Questions	• How do you find probability of an event? • How do you compare probabilities in different forms?
Lesson Narrative	In this lesson, students express the probability of events as fractions, decimals, and percentages. Students start with an Exploration on how to represent probability as a fraction, decimal, or percentage, relating these values ways of describing the likelihood of the event.

Component Summary				
	15.3.1 Warm-Up (~5 min)	Notice and Wonder about a set of cards (SE pg. 411)	15.3.4 Guided Instruction (10 min)	Expressing probability with fractions, decimals, and percentages (SE pg. 414)
	15.3.2 Exploration (10 min)	Explore probability in different forms (SE pg. 412)	15.3.5 Try It (10 min)	Practice writing and comparing probabilities (SE pg. 415)
	15.3.3 Bring It Together (5 min)	Summarize takeaways from Exploration (SE pg. 413)	15.3.6 Wrap-Up (~5 min)	Self-reflect on learning from the lesson (SE pg. 416)

Resources	Glossary	Materials	Additional Preparation
	Key Terms: • Probability Additional Terms: • Sample space • Chance	N/A	N/A

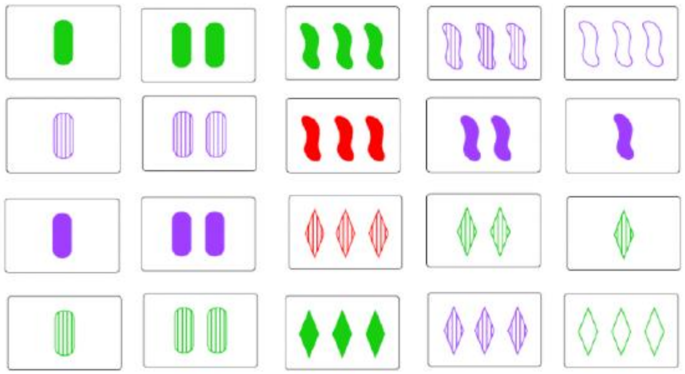
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may need support with converting between fractions, percentages, and decimals.
Students may struggle with the notation P(event) or mistake it as multiplication.</td><td>Consider ways to make connections to describing the likelihood of events based on their probability throughout the remainder of the lesson after this is addressed in the 15.3.2 Exploration and 15.3.3 Bring It Together.
Model how to read probability notation to ensure students understand.
Students may need reminders about what vowels and consonants are.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
In 15.3.5 Try It, consider implementing a Think-Pair-Share within question 1. This question contains a common misconception that students make when comparing probabilities. It is important to highlight this mistake.
Students can pair up with a partner and share their thoughts before coming together as a class to discuss.</td><td>MA.K12.MTR.2.1: Multiple Representations
Throughout this lesson, students learn to express probabilities in a variety of ways. They connect this variety of representations to descriptions of the likelihood of an event, building foundational understanding for the remainder of the unit.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">When first introduced, the format P(event) may be confusing to students. Consider reading these statements aloud in the 15.3.4 Guided Instruction to help students understand what probability they are finding.
i.e., P(vowel) – “What is the probability that a vowel will occur when Harper draws a letter out of the bag?”</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

15.3.1 Warm-Up: Notice and Wonder

Component Overview: *Students engage in a Notice and Wonder activity to activate the lesson.*

15.3.1 Warm-Up: Notice and Wonder

1. Some of the cards in the card game SET are shown.



a. What do you notice?

Answers will vary. I notice that the first three columns of cards have a pattern of increasing shapes on the cards by 1. I notice that there are less red than there are green and purple shapes.

b. What do you wonder?

Answers will vary. I wonder if there is a pattern with the rest of the cards in the card game.

Notice and Wonder
Consider having students share out what they noticed with the class.

- *If you were to select a card from those shown without looking, which number of shapes is the card most likely to have on it?*

15.3.2 Exploration: Representing Probabilities

Component Overview: *Students explore probability of events represented as a percentage, decimal, and fraction. Students are introduced to probability as it relates to the likelihood of an event happening and then explore what it means with a partner.*

15.3.2 Exploration: Representing Probabilities

When performing simple experiments, all of the outcomes have the same likelihood of happening. For example, when flipping a fair coin, the sample space includes two possible outcomes, $\{$heads, tails$\}$. Each outcome has an equal chance of happening. Therefore, the chance of landing on heads is $\frac{1}{2}$ because this event represents 1 outcome out of 2 possible outcomes. Represented as a decimal or a percentage, the chance of landing on heads is 0.5 or 50%, respectively.

Work with your partner to complete the following.

1. From the percentages listed below, determine the percent chance of each event occurring. Each value is used once.

   0% $16.\bar{6}\%$ $33.\bar{3}\%$ 50% 100%

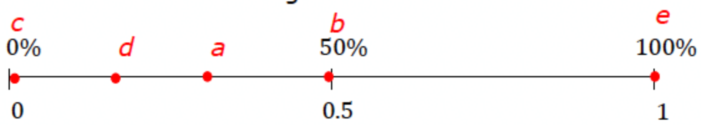
Event	Percent Chance
a. When given three doors with only one containing a prize behind it, what is the chance of opening the prize-winning door?	$33.\bar{3}\%$
b. When reaching into a dark closet with five pairs of shoes, what is the chance of pulling out a left shoe?	50%
c. When prompted to randomly select a crayon from a group of red, blue, and yellow crayons, what is the chance you choose a purple crayon?	0%
d. When rolling a fair six-sided die, what is the chance of landing on 1?	$16.\bar{6}\%$
e. When flipping a coin, what is the chance of landing on either heads or tails?	100%

Students may be tempted to write these as fractions instead of percentages.
Students can think about the values in fraction form first, and then convert that fraction to the closest percent.

Check for student understanding that the 3rd scenario is 0% and that the 5th scenario is 100%. These are important ideas for students to grasp.

15.3.2 Exploration: Representing Probabilities (continued)

2. Plot and label a point to represent the percent chance of each event happening from the previous question. Use the letters labeling each event.



The **probability** of an event is a measure of the likelihood the event will occur. Probabilities are expressed using numbers from 0 to 1 as a **decimal**, **percentage**, or **fraction**.

3. Consider the event from question 1 that is least likely to occur.

   a. Which event is the least likely to happen?
   Event c - choosing a purple crayon from a group of red, blue, and yellow crayons.

   b. Describe, using sentences, the likelihood of this event happening.
   Answers will vary. The likelihood of this event happening is impossible because there are 0 purple crayons to choose.

   c. Discuss with your partner whether or not it is possible to have a probability less than this.
   Answers will vary. It is not possible to have a probability less than this because impossible is the least likely an event can be.

- *What does it mean for the probability to be 0?*
- *What is the range of possible probabilities?*

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.
Listen as students discuss their thinking in response to this prompt to determine students' thought processes to highlight or misconceptions to address.

15.3.2 Exploration: Representing Probabilities (continued)

4. Consider the event from question 1 that is most likely to occur.

    a. Which event is the most likely to occur?
    Event e - flipping a coin and it landing on either heads or tails.

    b. Describe, in sentences, the likelihood of this event happening.
    Answers will vary. The likelihood of this event happening is certain because these are the only two possible outcomes. One or the other will always happen.

    c. Discuss with your partner whether or not it is possible to have a probability greater than this.
    Answers will vary. It is not possible to have a probability greater than this because 100% likelihood is the highest certainty.

5. Which event has the same likelihood of happening or not happening?
Event b - reaching into a dark closet with five pairs of shoes and pulling out a left shoe.

- *What does it mean to have a probability of 1?*

This is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference.
Listen as students discuss their thinking in response to this prompt to determine students' thought processes to highlight or misconceptions to address.

15.3.3 Bring It Together: Representing Probabilities

Component Overview: *Students summarize the concepts they explored in the previous component to draw conclusions about the probability scale and what it looks like in the form of a decimal, fraction, or percentage.*

15.3.3 Bring It Together: Representing Probabilities

1. Complete the statements.

The probability of an event can be any number between 0 and 1. A probability of 0 means it is

- certain
- impossible

for the event to occur. A probability of 1 means it is

- certain
- impossible

the event will occur. The closer a probability is to 1, the

- more
- less

likely it is to occur, while the closer it is to 0, the

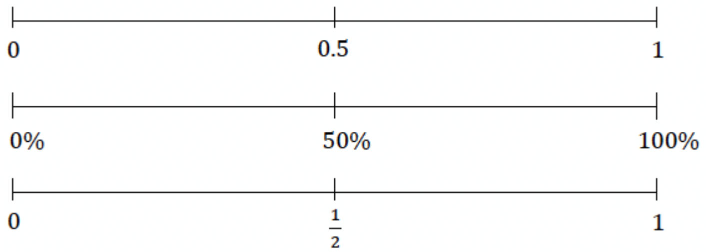
- more
- less

likely it is to occur. A probability of 0.5 means the event is

- likely.
- unlikely.
- neither likely nor unlikely.

Students should be able to work with their partner to complete question 1. The discussions in the previous component prime them for this summary. Review the number line together to ensure all students take away the necessary conclusions.

Probabilities can be represented by decimals, percentages, or fractions with equivalent values between 0 and 1.



To connect from question 1, write "certain" above 1, "impossible" above 0, and "neither likely nor unlikely" above 0.5 in the space above the number lines when concluding this component with students.

15.3.4 Guided Instruction: Expressing Probabilities of Events

Component Overview: *Students learn how to express the probability of an event, including the notation used. Students practice finding the probability of an event and converting the probability into different equivalent forms.*

15.3.4 Guided Instruction: Expressing Probabilities of Events

The **probability** of an event occurring, notated as $P(event)$, is determined by dividing the number of outcomes in the event by the possible outcomes in the sample space.

$$P(event) = \frac{\text{number of outcomes in the event}}{\text{total possible outcomes}}$$

Harper is playing a game where she draws a lettered tile out of a bag without looking.

The notation $P(\text{A})$ in this context means "the probability of pulling out a tile with the letter A on it."

1. Find the probability of each event, represented as a fraction.

   a. $P(\text{A})$
   $\frac{1}{11}$

   b. $P(\text{B})$
   $\frac{2}{11}$

   c. $P(\text{M})$
   $\frac{0}{11}$

   d. $P(\text{vowel})$
   $\frac{5}{11}$

   e. $P(\text{letter})$
   $\frac{11}{11}$

To extend students' thinking, consider having students draw a bag with the letters of their name in it.

- *What would be the probability of choosing each letter?*
- *Would any letters have a higher probability than others?*

This format P(A) may be confusing to students at first. Consider reading it aloud to help students understand what the question is asking them to do.
I.e., P(vowel) – "What is the probability that a vowel will occur when Harper draws a letter out of the bag?"

The probability of pulling a vowel or consonant is purposefully included in this guided component. These terms are often used in simple probability events, so this provides an opportunity to ensure students know the meaning of these words from ELA. Note that "y" is considered a vowel in this word.

15.3.4 Guided Instruction: Expressing Probabilities of Events (continued)

2. Consider the probability of pulling a tile with a consonant on it.

   a. Complete the table to find the fraction, decimal, and percentage equivalents of the probability.

Event	Fraction	Decimal	Percent
$P(\text{consonant})$	$\frac{6}{11}$	$0.\overline{54}$	$54.\overline{54}\%$

   The probability of this event is a repeating decimal. When this occurs, the decimal or percentage representation is typically rounded.

   b. What is $P(\text{consonant})$ represented as a decimal rounded to the nearest hundredth?
   0.55

   c. What is $P(\text{consonant})$ represented as a percentage rounded to the nearest tenth?
   54.5%

Prompt students to use the terms "likely" or "unlikely" when referring to the probabilities.
Consider asking the following to solidify understanding:

- *Which event from questions 1 or 2a is certain? Impossible? Unlikely? Somewhat likely?*

15.3.5 Try It: Expressing Probabilities of Events in Different Forms

Component Overview: *Students practice with expressing probabilities of events in different forms. Students also practice comparing probabilities in different forms, including fractions, decimals, and percentages.*

15.3.5 Try It: Expressing Probabilities of Events in Different Forms

1. The probability of rolling a 6 on a standard six-sided die is 16.67%, rounded to the nearest hundredth. Maisie says that this probability is greater than an event whose probability is 0.6 because $16.67 > 0.6$.

   Explain whether you agree or disagree with Maisie's conclusion. Support your argument by showing your work.
   Answers will vary. I disagree with Maisie because 16.67% as an equivalent decimal is $16.67\% \div 100 = 0.1667$, which is less than 0.6.

2. The probabilities of five events are given.

A	B	C	D	E
40%	0.25	$\frac{5}{6}$	42.9%	$\frac{1}{5}$

   a. Order the values of the probabilities from least to greatest.
   $\frac{1}{5}, 0.25, 40\%, 42.9\%, \frac{5}{6}$

Think-Pair-Share
Question 1 contains a common misconception that students make when comparing probabilities in different forms. Consider implementing a Think-Pair-Share within question 1. Students can pair up with a partner and share their thoughts before coming together as a class to discuss if needed.

- *How can you compare probabilities in different forms?*

Consider prompting students who are stuck to convert all probabilities to the same form so that they are easier to compare.

15.3.5 Try It: Expressing Probabilities of Events in Different Forms (continued)

b. Match the probabilities with the events described in the table.

Event	Probability
 Landing on green on the spinner shown	25%
Pulling a blue marble from a bag containing equal numbers of red, orange, yellow, green, and blue marbles	$\frac{1}{5}$
Landing on a factor of 10 when rolling a fair ten-sided die	40%
Picking a card with a vowel on it from a set of cards with each letter in the word FLORIDA on it	42.9%
Landing on a number less than 6 on a standard six-sided die	$\frac{5}{6}$

c. Explain which event is the least likely to occur based on the probabilities.
Answers will vary. The event that is least likely to occur is pulling a blue marble from a bag containing equal numbers of red, orange, yellow, green, and blue marbles because it has the smallest probability.

Notice which students are having a hard time comparing probabilities. Consider using this information for differentiation strategies or groupings in the next lesson, as students will be using this skill.

15.3.6 Wrap-Up: Reflection

Component Overview: *Students reflect on their current level of understanding based on this lesson.*

15.3.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current level of understanding is related to each target.

1. I can express the probability of a chance event using a fraction, percentage, or decimal.
   Answers will vary.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.
		X

2. I can compare probabilities of chance events.
   Answers will vary.

I need help. I can't get started.	I'm getting there, but I need more practice.	I understand this and I feel confident.
		X

The information provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

15.4 Comparing Probabilities

<table>
<tr><th colspan="5">Lesson Introduction</th></tr>
<tr><td>Benchmark</td><td colspan="2">MA.7.DP.2.2 Given the probability of a chance event, interpret the likelihood of it occurring. Compare the probabilities of chance events.</td><td>Learning Targets</td><td>I can interpret the likelihood of a chance event occurring using probability.
I can compare probabilities of chance events.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="2">Building On</td><td colspan="2">Working Toward</td></tr>
<tr><td colspan="2">N/A</td><td colspan="2">N/A</td></tr>
<tr><td>Essential Questions</td><td colspan="4">How can you express the likelihood of a chance event as a probability?
How can you compare probabilities expressed in different forms?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="4">In this lesson, students will practice determining probability of an event, describing its likelihood, and comparing it to the probabilities of other events. Students will complete a Card Sort comparing and describing probabilities of chance events before practicing on their own. To conclude the lesson, students compare two student responses when interpreting and comparing probabilities of two events.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>15.4.1 Warm-Up (~5 min)</td><td>Career Video – Statistician (SE pg. 417)</td><td>15.4.3 Guided Practice (10 min)</td><td>Practice interpreting and comparing probabilities (SE pg. 418)</td></tr>
<tr><td>15.4.2 Activity (15 min)</td><td>Probability Card Sort (SE pg. 417)</td><td>15.4.4 Your Turn! (5–10 min)</td><td>Practice ordering probabilities in different forms (SE pg. 419)</td></tr>
<tr><td>15.4.5 Wrap-Up (~5 min)</td><td colspan="3">Compare two student responses to determine who is correct (SE pg. 420)</td></tr>
<tr><td rowspan="2">Resources</td><td>Glossary</td><td colspan="2">Materials</td><td>Additional Preparation</td></tr>
<tr><td><u>Key Terms:</u> N/A

<u>Additional Terms:</u>
Sample space
Chance
Probability
Likelihood</td><td colspan="2">Career Profile Video
Cards for 15.4.2 Activity</td><td>15.4.1 Warm-Up contains a video to accompany the question prompts.</td></tr>
</table>

Teacher Tips

Common Misconceptions	Things to Consider
• Students may get confused with probability and likelihood. The scaffolding in this lesson first has students order events by probability and then describe the likelihood. • Students may need support with converting between fractions, percentages, and decimals.	• Consider ways to make connections among describing the likelihood of events based on their probability throughout the lesson.

Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
Think-Pair-Share In the 15.4.2 Activity, after students order their first set of cards with their partner, students can compare/discuss with another group to see how they ordered their cards before continuing with the next set of cards.	**MA.K12.MTR.7.1: Real-World Contexts** Students may have experience with seeing statisticians in the real world, like watching sports and hearing commentators share statistics about players or teams. Statistics have many applications in a variety of career fields that are not typically thought of as STEM. Use 15.4.1 Warm-Up as an opportunity to highlight this and the importance of understanding statistics in the real world to avoid manipulation.

Differentiate Instruction

Consider a kinesthetic approach to 15.4.3 question 2. Read the problem aloud and have students move to one side of the room or the other depending on which statement they think is more likely. Students who think the statements are equally likely can stand in the middle of the room.

Lesson Notes:

15.4.1 Warm-Up: Statisticians Careers

Component Overview: *Play the Career Profile video on the Statistician for the entire class. After students watch the video (2:06 minutes), prompt them to reflect on the profession by answering the questions independently.*

15.4.1 Warm-Up: Statisticians Careers

Statisticians turn data into understandable and useful information. The government, health care industry, sports broadcasting, and many businesses have statisticians who analyze data to predict trends. Statisticians help businesses and industries with decision-making based on analysis.

1. What are some real-world examples where you have seen statistics used?
   Answers will vary.

2. Consider an industry you are interested in working in someday.

   a. What industry are you considering?
      Answers will vary.

   b. What is at least one way you could see trends in data and analysis of statistics used in this industry?
      Answers will vary.

MA.K12.MTR.7.1: Mathematics in the Real World
Students may have experience with seeing statisticians in the real world, like watching sports and hearing commentators share statistics about players or teams. Statistics has many applications in a variety of career fields that are not typically thought of as STEM. Use this Warm-Up as an opportunity to highlight this and the importance of understanding statistics in the real world to avoid manipulation.

15.4.2 Activity: Likelihood of Chance Events

Component Overview: *Students will work with a partner to order the events on the Card Sort from least likely to most likely. After sorting the cards, students work with their partners to answer questions where they describe the likelihood of events or express them using probabilities.*

15.4.2 Activity: Likelihood of Chance Events

Work with your partner or group to order the events on the cards from least likely to most likely.

After ordering the events, work with your partner or group to complete the following using the events from the card sort.

1. Explain which event(s), if any, can be described as "impossible."
   Answers will vary. These will be the events that have a 0% chance of happening.
   - *Drawing out a ball with the number 3 from a fishbowl containing balls with even numbers written on them*

2. Explain which event(s), if any, can be described as "certain."
   Answers will vary. These will be the events that have a 100% chance of happening.
   - *Choosing a block with 4 sides of the same length from a pile of square blocks*

3. Explain which event(s), if any, can be described as "equally likely as not."
   Answers will vary. These will be any event that has a 50% chance of happening or not.
   - *Pulling a red card from a deck with half the cards red*

4. Express the likelihood of a randomly chosen word in a novel beginning with the letter T, as a percentage.
   $\frac{4}{25} = 0.16; \quad 0.16 \cdot 100 = 16\%$

5. Express the likelihood of a randomly selected person being left-handed, as a decimal.
   $10\% = \frac{10}{100} = 0.1$

Consider using the two sets of cards separately so that you can monitor for understanding and address any common misconceptions after the first set before giving the next set.

Fast finishers can go through and express the probability of each even number in the Card Sort.

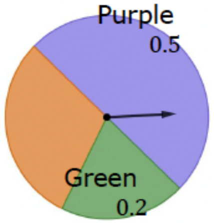
15.4.3 Guided Practice: Interpreting and Comparing Probabilities

Component Overview: *Students practice interpreting and comparing probabilities of chance events.*

15.4.3 Guided Practice: Interpreting and Comparing Probabilities

1. A spinner is shown with the probabilities of landing on each space given as a decimal.

   Use the spinner to complete the table by identifying the missing events, probabilities, and descriptions of likelihood.



Event	Probability	Likelihood
$P(\text{purple})$	50%	o impossible o unlikely o equally likely as not o likely o certain
$P(\text{orange})$	30%	o impossible o unlikely o equally likely as not o likely o certain
$P(\text{yellow})$	0%	o impossible o unlikely o equally likely as not o likely o certain
$P(\text{green})$	20%	o impossible o unlikely o equally likely as not o likely o certain
$P(\text{purple or orange})$	80%	o impossible o unlikely o equally likely as not o likely o certain
$P(\text{color})$	100 %	o impossible o unlikely o equally likely as not o likely o certain

Consider prompting students to discuss the possible outcomes of spinning the spinner, including the likelihood of landing on each color prior to having them engage with the table on the next page.

Students can create their own spinner and table and determine the probability and likelihood of the events in the spinner.

15.4.3 Guided Practice: Interpreting and Comparing Probabilities (continued)

2. For each scenario, circle the event that is more likely. If the events are equally likely, circle both.

   a. Alex and Elvis play baseball. Based on their statistics, Alex's probability of hitting a home run is 0.27. The probability of Elvis hitting a home run is 14%. Which choice is more likely on their next at bat?

Alex hits a home run.	or	Elvis hits a home run.

   b. A meteorologist states there is a 50% chance for rain tomorrow. Which choice is more likely to happen?

It will rain tomorrow.	or	It will not rain tomorrow.

   c. Shawn is calling the coin toss before his football game. Which choice is more likely to occur?

The coin will land on its edge.	or	The coin will land on either heads or tails.

Consider a kinesthetic approach to this question. Read the question aloud and have students move to one side of the room or the other depending on which statement they think is more likely. If they think they are equal, they can stand in the middle of the room.

15.4.4 Your Turn!

Component Overview: *Students practice comparing and ordering the probabilities of chance events expressed in different forms.*

15.4.4 Your Turn!

1. The probability that Lawrence makes a free throw when playing basketball is 60%. The likelihood that he makes a 3-point shot is 0.345.

   Explain which event is more likely, Lawrence making a free throw or making a 3-point shot.

   The event that is more likely is that Lawrence will make a free throw because the likelihood is 60%. Converting the probability that he will make a 3-pointer (0.345) to a percentage is 34.5% which is less the 60% for the free throw.

2. The probabilities of five events are shown.

   60% $\quad$ 8 out of 10 $\quad$ 0.37 $\quad$ 20% $\quad$ $\frac{5}{6}$

   List the probabilities in order from least to greatest.

   $20\%,\quad 0.37,\quad 60\%,\quad 8\ out\ of\ 10,\quad \frac{5}{6}$

How can you compare probabilities in different forms?

15.4.5 Wrap-Up: Ticket Out the Door

Component Overview: *Students compare two student responses to determine who is correct when comparing the likelihood of two different events.*

15.4.5 Wrap-Up: Ticket Out the Door

1. There are 25 prime numbers between 1 and 100. There are 46 prime numbers between 1 and 200. Two events are shown.

Event A	Event B
A computer produces a random number between 1 and 100 that is prime.	A computer produces a random number between 1 and 200 that is prime.

Ava said Event B is more likely to occur because there are 46 possible outcomes in the event compared to only 25 in Event A. Oscar said Event A is more likely because $P(\text{A}) = 0.25$ and $P(\text{B}) = 0.23$.

a. Explain who is correct.
Oscar is correct because the probability for Event A is $P(A) = 0.25$ because $\frac{25}{100} = 0.25$ and the probability for event b is $P(B) = 0.23$ because $\frac{46}{200} = 0.23$.

b. Write a short note to the student who was incorrect explaining the error they made.
Answers will vary. Ava, probability is the ratio of favorable outcomes to total outcomes. It is not enough to simply compare the number of favorable outcomes without comparing to the total possible outcomes.

The responses provided from the student here, coupled with what was observed during instruction, can provide information for differentiating instruction in later lessons.

15.5 Experimental and Theoretical Probability

Lesson Introduction

Benchmarks	MA.7.MA.DP.2.3 Find the theoretical probability of an event related to a simple experiment. MA.7.MA.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.	Learning Targets	• I can perform a simulation to determine an experimental probability. • I can find the theoretical probability of an event. • I can compare the experimental probability from a simulation to the theoretical probability of a simple experiment.

Benchmark Coherence	Building On	Working Toward
	N/A	MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment. MA.8.DP.2.3 Solve real-world problems involving probabilities related to single or repeated experiments, including making predictions based on theoretical probability.

Essential Questions	• What is the difference between theoretical and experimental probability? • How do you determine the theoretical and experimental probabilities of a simple experiment?
Lesson Narrative	In this lesson students are introduced to the terms “theoretical probability” and “experimental probability” and given opportunities to explore the difference. Students compare the theoretical and experimental probabilities of a simple experiment over different numbers of trials to build a strong foundational understanding of the difference between the two concepts. Students also begin exploring how the number of trials impacts the closeness of the experimental probability to the theoretical probability, a concept that will be further developed in Lesson 6.

Component Summary				
	15.5.1 Warm-Up (~5 min)	Determine equivalent probabilities (SE pg. 421)	15.5.4 Guided Instruction (5 min)	Introduce theoretical and experimental probability (SE pg. 425)
	15.5.2 Exploration (15-20 min)	Simulate a simple experiment (SE pg. 421)	15.5.5 Try It (5-10 min)	Practice interpreting theoretical and experimental probability in context (SE pg. 425)
	15.5.3 Bring It Together (10 min)	Observe patterns after simulating a simple experiment (SE pg. 423)	15.5.6 Wrap-Up (~5 min)	Summarize the difference between theoretical and experimental probability (SE pg. 426)

Resources	Glossary	Materials	Additional Preparation
	<u>Key Terms:</u> • Theoretical probability • Experimental probability <u>Additional Terms:</u> • Probability	• 6-sided number cube • Calculators	• This is a vocabulary-rich lesson. Consider preparing your classroom-specific vocabulary or glossary strategies for support (Word Wall, Math Cards, etc.)

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may confuse theoretical and experimental probabilities. • Students may make mistakes when converting from a fraction to a decimal.	• Consider having students summarize the definitions of theoretical and experimental probability in their own words through a Turn-and-Talk during the 15.5.4 Guided Instruction.
	Literacy and Language Routines	**Mathematical Thinking and Reasoning (MTR) Standard**
	Notice and Wonder In Exploration 15.5.2 have students engage in a Notice and Wonder. What do you notice about the points on the graph? What do you wonder? As the number of rolls increases, what do you notice happening on the graph?	**MA.K12.MTR.4.1: Discussion and Reflection** This lesson creates many opportunities for students to discuss their thinking with peers. In 15.5.3 Bring It Together, students have a chance to engage in discourse as a class regarding theoretical vs. experimental probability.
Differentiate Instruction	In the 15.5.4 Guided Instruction, the definitions of theoretical and experimental probability are wordy. Have students go through and highlight or underline important words. Students can each come up with a definition for theoretical and experimental probability in their own words.	

Lesson Notes:

15.5.1 Warm-Up: Marbles

Component Overview: *Students complete two statements that represent equivalent probabilities in the event of drawing a red marble from a bag.*

15.5.1 Warm-Up: Marbles

1. Fill in the blanks in the statements below so the probability of drawing a red marble from either bag is the same. Use any number 1 through 9 at most one time each.
   Answers will vary.

   There are 2 red marbles and 4 blue marbles in Bag A.

   There are 3 red marbles and 6 green marbles in Bag B.

- *How can you show that the probabilities are equivalent?*
- *What is the probability expressed as a percentage?*

15.5.2 Exploration: Playing the Long Game

Component Overview: *In this Exploration, students explore experimental probability for the first time. They start by coming up with the theoretical probability first and then complete the experiment while recording their results. Students compare the theoretical and experimental probabilities.*

15.5.2 Exploration: Playing the Long Game

Samaria plays a game with a fair six-sided die in which she only wins if she rolls a 1 or a 2.

1. List the outcomes in the sample space for rolling the fair die.

   $1, 2, 3, 4, 5,$ *or* 6

2. Discuss the probability of Samaria winning the game with your partner or group. Then, complete the statement.

   The probability Samaria wins the game, represented as a fraction, is $\frac{2}{6}$ *or* $\frac{1}{3}$ because there are 2 possible winning outcomes out of 6 total possible outcomes from rolling the die.

3. If Samaria is given the option to flip a coin and win if it comes up heads, explain whether or not that would be a better option for her to win using probability.

   This would be a better option because the probability with the dice is $\frac{2}{6}$ *or* 33.3% *and the probability for a coin flip landing heads up is* $\frac{1}{2}$ *or* 50%

This Exploration can be done in partners or groups of 3-4. Data is gathered from each group and used in the 15.5.3 Bring It Together, so having fewer groups may make gathering the data easier.

15.5.2 Exploration: Playing the Long Game (continued)

4. With your partner or group, follow these instructions 10 times to create the graph.
   - One person rolls the number cube. Everyone records the outcome.
   - Calculate the fraction of all rolls that are wins for Samaria so far. Approximate the fraction with a decimal value rounded to the hundredths place using a calculator. Record both the fraction and the decimal in the last column of the table.
   - On the graph, plot the number of rolls and the fraction of rolls that were wins.
   - Pass the number cube to the next person in the group.

Answers will vary.

Roll	Outcome	Total Number of Wins for Samaria	Fraction of Games Played that are Wins
1	3	0	$\frac{0}{1} = 0.0$
2	2	1	$\frac{1}{2} = 0.5$
3	3	1	$\frac{1}{3} = 0.33$
4	4	1	$\frac{1}{4} = 0.25$
5	5	1	$\frac{1}{5} = 0.2$
6	1	2	$\frac{2}{6} = 0.33$
7	1	3	$\frac{3}{7} = 0.43$
8	3	3	$\frac{3}{8} = 0.38$
9	5	3	$\frac{3}{9} = 0.33$
10	4	3	$\frac{3}{10} = 0.3$

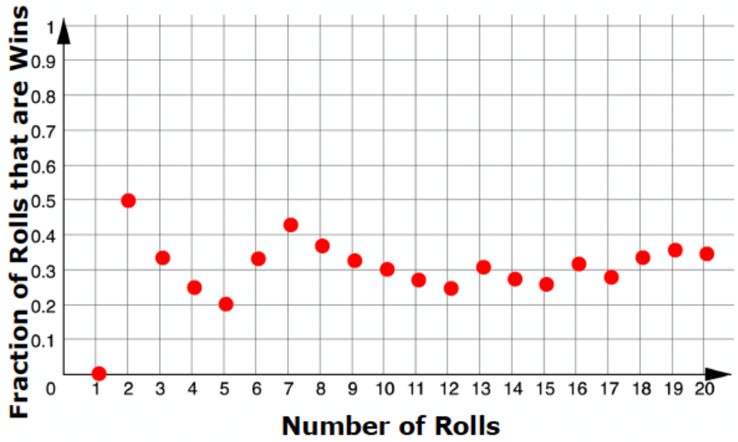
What is the likelihood of Samaria winning the game?

If you do not have a number cube, cut pieces of paper labeled 1-6 and randomly choose with replacement.

Consider going over how to find the fraction of games played that are wins. Consider posting this on the white board to help.

$$\frac{number\ of\ wins}{number\ of\ trials}$$

15.5.2 Exploration: Playing the Long Game (continued)



5. With your partner or group, discuss what you notice is happening with the points on the graph.
   Answers will vary.
   a. After 10 rolls, what fraction of the total rolls were a win?
      Answers will vary. In the example shown the fraction of rolls that were a win is $\frac{3}{10}$.

   b. How close is this fraction to the probability that Samaria will win?
      Answers will vary.

Notice and Wonder

- *What do you notice about the points on the graph?*
- *What do you wonder?*
- *As the number of rolls increases, what do you notice happening on the graph?*

15.5.2 Exploration: Playing the Long Game (continued)

6. Roll the number cube 10 more times. Record your results in this table and add them to the graph from earlier.
   Answers will vary.

Roll	Outcome	Total Number of Wins for Samaria	Fraction of Games Played that are Wins
11	6	3	$\frac{3}{11} = 0.27$
12	6	3	$\frac{3}{12} = 0.25$
13	2	4	$\frac{4}{13} = 0.31$
14	3	4	$\frac{4}{14} = 0.29$
15	3	4	$\frac{4}{15} = 0.27$
16	1	5	$\frac{5}{16} = 0.31$
17	3	5	$\frac{5}{17} = 0.29$
18	1	6	$\frac{6}{18} = 0.33$
19	1	7	$\frac{7}{19} = 0.37$
20	4	7	$\frac{7}{20} = 0.35$

7. Observe any trends on the graph after 20 rolls.

   a. After 20 rolls, what fraction of the total rolls was a win?
      Answers will vary. In the example shown $\frac{7}{20}$ rolls resulted in a win.

   b. How close is this fraction to the probability that Samaria will win?
      Answers will vary. It is close to the probability of Samaria winning, $\frac{2}{6}$, or $\frac{1}{3}$.

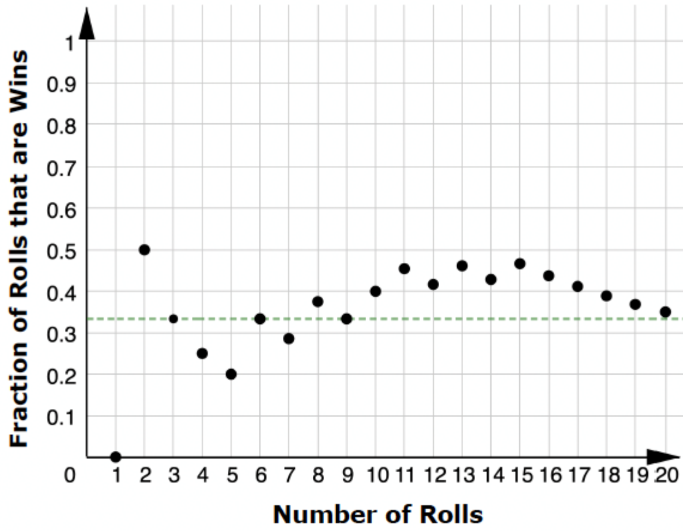
Notice and Wonder
- *What do you notice about the table?*
- *What do you wonder?*
- *As the number of rolls increases, what do you notice happening in the table?*
- *How does this relate to what is happening in the graph?*

15.5.3 Bring It Together: Playing the Long Game

Component Overview: *As a class, students will look at the data they have collected and compare it to the probability that Samaria would win the game.*

15.5.3 Bring It Together: Playing the Long Game

A sample graph from a group who performed the same simple experiment in the previous activity is shown.



1. What does the green dotted line represent on the graph?
   The dotted line represents the probability of rolling a 1 or a 2, $\frac{2}{6}$, or $\frac{1}{3}$

Students may think the dotted line represents an average probability, rather than the probability of winning.

2. Consider the data from all of the groups performing the experiment.

   a. Complete the table below with your class by adding the overall data from each group. The first row has already been completed using the sample data above.
   Answers will vary.

Group	Total Number of Wins for Samaria	Fraction of Games Played that are Wins
Sample Data	7	$\frac{7}{20} = 0.35$

   b. Write the fraction that represents the total number of wins out of the total number of games played by all groups.
   Answers will vary.

   c. How close is this fraction to the probability that Samaria will win?
   Answers will vary.

- *Do you think the number of trials impacts how close the total number of wins out of the total number of games played is to the predicted probability of winning?*
- *Do you think that if you conducted this experiment again, you would get the exact same results?*

The probability for an event represents the proportion of the time it is expected for the event to occur in the long run, over many trials. It does not mean that is exactly how many times the event will actually occur.

15.5.4 Guided Instruction: Experimental and Theoretical Probability

Component Overview: *Students are introduced to the definitions of theoretical and experimental probability before summarizing the Exploration activity using these terms.*

15.5.4 Guided Instruction: Experimental and Theoretical Probability

Consider the terms defined below and how they relate to the previous Exploration.

A ***theoretical probability*** is a number between 0 and 1 that represents the likelihood of an event in a theoretical model based on a sample space. If all outcomes in the sample space are equally likely, then the theoretical probability of an event is the ratio of the number of outcomes in the event to the number of outcomes in the sample space.

An ***experimental probability*** is a number between 0 and 1 that shows the ratio of the number of times an event occurs to the total number of trials or times the activity is performed.

1. Use the definitions to complete the statements.

   In the Exploration, the [○ experimental / ○ theoretical] probability of Samaria winning the game was $\frac{1}{3}$ because there were [○ 2 / ○ 3 / ○ 6] possible outcomes out of [○ 2 / ○ 3 / ○ 6] in the sample space that won. By perfoming the activity, each group determined [○ an experimental / ○ a theoretical] probability for the experiment.

Highlight or underline important words in the definition. Consider giving students the opportunity to turn to a partner and share what they understand about each term in their own words.

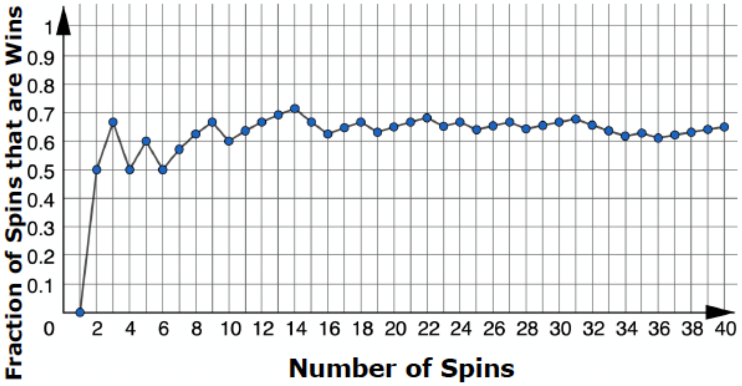
15.5.5 Try It: Experimental and Theoretical Probability

Component Overview: *Students apply what they have learned about experimental and theoretical probability. Students should be able to determine the difference between which probabilities are being found.*

15.5.5 Try It: Experimental and Theoretical Probability

1. Deepa wants to know if flipping a quarter really does have a probability of $\frac{1}{2}$ of landing heads-up, so she flips a quarter 10 times. It lands heads-up 3 times and tails-up 7 times. Explain whether or not she has proven that the probability of landing heads up is not $\frac{1}{2}$.
Deepa has found the experimental probability, this may not be exactly $\frac{1}{2}$. If she continues the trials over and over, the probability average will get closer and closer to $\frac{1}{2}$.

2. A spinner is spun 40 times for a game. The graph shows the fraction of games that are wins.



   a. Estimate the probability of a spin winning this game based on the graph.
   Answers will vary but should be around 0.6 *or* 60%

   b. What type of probability is your estimate?
   Experimental Probability.

Consider asking students the following:

- *How many trials did Deepa conduct?*
- *Do you think that was enough trials?*
- *Do you think an experimental probability being different from the theoretical probability means that the theoretical probability was wrong?*

15.5.6 Wrap-Up: Cool Down

Component Overview: *Students summarize what they have learned today with a quick note to a friend.*

15.5.6 Wrap-Up: Cool Down

1. Your friend was absent today and will need the notes from class. Write a quick note to help them understand the difference between the theoretical and experimental probability of a simple experiment.
   Answers will vary.

Consider using data gathered from this Wrap-Up to inform differentiation strategies in Lessons 6 and 7 as students will be applying experimental and theoretical probability in the following lessons.

15.6 Simulating Small and Large Numbers of Trials

Lesson Introduction

Benchmark	MA.7.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.	Learning Targets	• I can compare the experimental probability from a simulation to the theoretical probability of a simple experiment. • I can use results from computer-simulated events to informally define the Law of Large Numbers.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment. MA.8.DP.2.3 Solve real-world problems involving probabilities related to single or repeated experiments, including making predictions based on theoretical probability.

Essential Questions	• How can you use computer-simulated events to compare experimental and theoretical probabilities? • How does the number of trials in a simple experiment affect the experimental probability?
Lesson Narrative	This lesson is designed to give students an opportunity to use simulations to collect data, then calculate experimental probabilities and estimate theoretical probability of an event using the experimental probabilities they calculate. Different methods for conducting simulations are offered in the lesson. Some simulations use technology, while others do not. Students then practice matching situations to a simulation that could be used to determine probabilities of events.

Component Summary	15.6.1 Warm-Up (~5 min)	Which spinner doesn't belong? (SE pg. 427)	15.6.3 Bring It Together (10 min)	Choosing a simulation for a situation (SE pg. 429)
	15.6.2 Exploration (15–20 min)	Compare experimental results from a simulation to expected results (SE pg. 427)	15.6.4 Try It (10 min)	Matching situations with simulations to determine experimental probabilities (SE pg. 430)
	15.6.5 Wrap-Up (~5 min)	Self-reflection on the lesson (SE pg. 431)		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms:</u> • Simulation <u>Additional Terms:</u> • Random variation • Experimental probability • Theoretical probability	• N/A	• (Optional) 15.6.4 Try It can be turned into task cards.

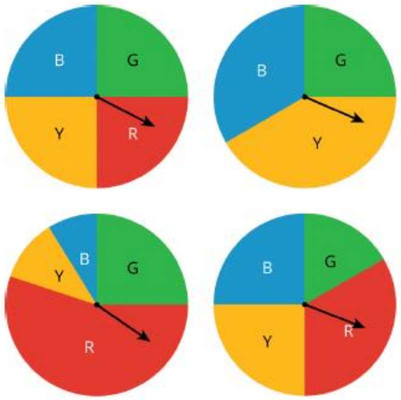
<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students may confuse experimental and theoretical probabilities.
Students may struggle with matching a situation with a simulation that could be used to determine the experimental probability.</td><td>Consider turning the 15.6.4 Try It into a matching activity where students physically match the cards.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Think-Pair-Share
In 15.6.2 Exploration, question 2c lends itself to have students engage in a Think-Pair-Share. Consider having students write down a few notes about what they notice before pairing up and comparing ideas. Partners can then share out in a class discussion what they noticed.</td><td>MA.K12.MTR.4.1: Discussion and Reflection
This lesson creates many opportunities for students to discuss their thinking with peers. In 15.6.4 Try It, students work together to match the situation with the simulation that would best determine the experimental probabilities.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">For a kinesthetic approach to 15.6.4 Try it, consider creating task cards out of the descriptions and simulations. Students can then physically match the descriptions to the simulations with their partner.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

15.6.1 Warm-Up: Which One Doesn't Belong?

Component Overview: *Students are presented with four spinners with the prompt "Which one doesn't belong?" Students could argue that any one of the four options "doesn't belong" for different reasons using mathematical reasoning.*

15.6.1 Warm-Up: Which One Doesn't Belong?

1. Four spinners are shown.



a. Which spinner doesn't belong?
Answers will vary. Sample answers:
- *The top right spinner doesn't have red.*
- *The bottom left spinner is the only one with red > 50%.*
- *The top left spinner is the only fair spinner.*
- *The bottom right spinner is the only one that the green sector doesn't have a 25% of being spun.*

b. Explain your reasoning.
Answers will vary.

"Which One Doesn't Belong?" fosters a need to define terms carefully and use words precisely (probability, experiment, trials, outcome, sample space, simulation ...) in order to compare and contrast the group of spinners. Consider prompting students to include at least one term learned in this unit in their reasoning.

15.6.2 Exploration: Comparing Experimental Results to Expected Results

Component Overview: *Students engage in an Exploration where they look at the results of different numbers of trials on a simulation of spinning a color spinner. Students start by finding the theoretical probability of landing on each color and compare that to the experimental probabilities from the simulation.*

15.6.2 Exploration: Comparing Experimental Results to Expected Results

Work with your partner to complete the following.

1. When performing the simple experiment of flipping a coin over repeated trials, imagine each of the resulting situations listed in the table.

   a. Complete the table to describe the results.

Situation	Is the result surprising?		Is the result possible?	
	Yes	No	Yes	No
You flip the coin once, and it lands heads-up.	○	●	●	○
You flip the coin twice, and it lands heads-up both times.	○	●	●	○
You flip the coin 100 times, and it lands heads-up all 100 times.	●	○	●	○

   b. If you flip the coin 100 times, explain how many times you would expect the coin to land heads-up.
   Answers will vary. But you would expect about half of the trials to land face up.

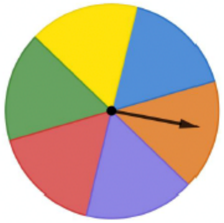
   c. With your partner, discuss some other results that would not be surprising if you flip the coin 100 times.
   Answers will vary. Answers will be around the number 50.

- *Would you expect to get the exact same result each time?*
- *Why or why not?*

Students can use the simulation applet for flipping a coin.

15.6.2 Exploration: Comparing Experimental Results to Expected Results (continued)

2. A spinner is shown.



a. Find the theoretical probability of landing on each color as a percentage rounded to the nearest tenth.

Color	Red	Purple	Orange	Blue	Yellow	Green
Theoretical Probability	16.7%	16.7%	16.7%	16.7%	16.7%	16.7%

b. Sara used a computer-simulated spinner to perform different numbers of trials with the spinner. The results show the percentage of times it landed on each color in each simulation.

Circle the experimental probabilities that are more than 1% different from the theoretical probability.

Color	Red	Purple	Orange	Blue	Yellow	Green
Simulation A (6 spins)	0%	16.7%	16.7%	33.3%	0%	33.3%
Simulation B (60 spins)	16.7%	15%	20%	13.3%	10%	25%
Simulation C (600 spins)	17.7%	19%	18%	15.8%	14.5%	15%
Simulation D (6000 spins)	16.9%	16%	17.1%	16.6%	16.7%	17.1%

Consider a reminder of how to find theoretical probability.

$$\frac{\text{number of desired outcomes}}{\text{total number of possible outcomes}}$$

Consider posting this formula on the board for students to reference.

- *Is this simulation method free of bias?*

Notice and Wonder
- *What do you notice about how many experimental probabilities you circled in each simulation?*
- *What do you wonder about why this is?*

15.6.2 Exploration: Comparing Experimental Results to Expected Results (continued)

c. Discuss with your partner what you notice when comparing the experimental probabilities to the theoretical probabilities.
Answers will vary.

d. In which simulation(s) were the experimental probabilities the most different from the theoretical probabilities?
Simulation A was most different from the theoretical probability.

e. In which simulation(s) were the experimental probabilities the most similar to the theoretical probabilities?
Simulation D was most similar to the theoretical probability.

f. Imagine Sara used the computer-simulated spinner to perform 600,000 spins. Describe how you think the experimental probabilities of landing on each color would compare to the theoretical probabilities.
Answers will vary. The experimental probability should end up being close to the theoretical probability the more times that the spinner is spun.

Think-Pair-Share
Question 2c lends itself to have students engage in a Think-Pair-Share. Consider having students write down a few notes about what they notice before pairing up and comparing ideas. Partners can then share out in a class discussion what was noticed.

- *Why do you think this is?*

15.6.3 Bring It Together: Comparing Experimental Results to Expected Results

Component Overview: *In this Bring It Together, students continue to look at simulation and experimental results of a simulation. Students also look at how they can determine a simulation that could be used to predict results for various chance events.*

15.6.3 Bring It Together: Comparing Experimental Results to Expected Results

A ***simulation*** is an approximate imitation of a statistical experiment, often done with a computer program to examine the statistics of a large quantity of trials.

1. Complete the statement.

   As the number of trials in a simulation of a simple experiment increases, the [○ experimental / ○ theoretical] probability gets [○ closer to / ○ farther from] the [○ experimental / ○ theoretical] probability.

2. Imagine flipping a coin 3 times, and it has come up heads once. The experimental probability of landing on heads after 3 trials is $\frac{1}{3}$. Discuss with your partner whether flipping the coin one more time is guaranteed to result in landing on heads so the experimental probability, $\frac{2}{4}$, will equal the theoretical probability $\frac{1}{2}$.

   Answers will vary. It is not a guarantee that the coin will land on heads, but the more times that the coin is flipped, the closer the experimental probability will get to the theoretical probability.

Listen to student discussions to determine their understanding of random variation.
When debriefing this question, call on different students to share what they discussed with their partner based on what they heard.
The goal is to lead the debrief toward understanding that on each trial, the probability of landing on heads is $\frac{1}{2}$ but that due to random variation, it doesn't always work out that the coin lands on heads exactly half the time.

15.6.3 Bring It Together: Comparing Experimental Results to Expected Results (continued)

When performing simple experiments, the experimental probabilities may differ from theoretical probabilities due to random variation. However, as the number of repetitions increases, experimental probabilities will typically better approximate the theoretical probabilities.

Simulations are useful when it is hard or time consuming to gather enough information to estimate the probability of some event.

3. Consider the situations described in the table.

Situation	Description
A	The local meteorologist in Ocala, FL says there is a 50% chance of rain tomorrow.
B	A science teacher at Hamilton County High School puts the names of all students in a bag and randomly selects which of 6 groups they will sit in for his new seating chart.
C	The U.S. Centers for Disease Control and Prevention (CDC) released data in 2016 showing that $\frac{1}{3}$ of all adults in the US do not get enough sleep on a regular basis.

Explain which situation could be simulated using the spinner from the Exploration.

The situation that could best be used by a spinner simulation is Situation B. The spinner has 6 sections, and the science teacher has 6 groups. The spinner will provide a truly random selection.

Answers may vary as students could also make an argument for Situation A.

Consider a class discussion on what random variation means in an experiment.

Consider asking students to highlight or underline any words or phrases which they aren't sure how to explain. This can provide insight into vocabulary or contexts that may need to be discussed to build students' understanding.

15.6.4 Try It: Using Simulations to Determine Experimental Probabilities

Component Overview: *In this Try It activity, students match four descriptions of situations to a simulation that could be used to determine the experimental probabilities of the event.*

15.6.4 Try It: Using Simulations to Determine Experimental Probabilities

1. Four situations are described.

Situation	Description
A	In a small lake, 25% of the fish are female. You capture a fish, record whether it is male or female, and toss the fish back into the lake. If you repeat this process 5 times, what is the probability that at least 3 of the 5 fish are female?
B	Elena makes about 80% of her free throws. Based on her past successes with free throws, what is the probability that she will make exactly 4 out of 5 free throws in her next basketball game?
C	On a game show, a contestant must pick one of three doors. In the first round, the winning door has a vacation. In the second round, the winning door has a car. What is the probability of winning a vacation and a car?
D	Your choir is singing in 4 concerts. You and one of your classmates both learned the solo. Before each concert, there is an equal chance the choir director will select you or the other student to sing the solo. What is the probability that you will be selected to sing the solo in exactly 3 of the 4 concerts?

Work with your partner to match each situation to a simulation on the next page.

For a kinesthetic approach to this activity, consider creating task cards out of the descriptions and simulations. Students can then physically match the descriptions to the simulations with their partner.

15.6.4 Try It: Using Simulations to Determine Experimental Probabilities (continued)

Simulation	Situation
Toss a standard number cube 2 times and record the outcomes. Repeat this process many times and find the proportion of the simulations in which a 1 or 2 appeared both times to estimate the probability.	*C*
Make a spinner with four equal sections labeled $1, 2, 3$, and 4. Spin the spinner 5 times and record the outcomes. Repeat this process many times and find the proportion of the simulations in which a 4 appears 3 or more times to estimate the probability.	*A*
Toss a fair coin 4 times and record the outcomes. Repeat this process many times, and find the proportion of the simulations in which exactly 3 heads appear to estimate the probability.	*D*
Place 8 blue chips and 2 red chips in a bag. Shake the bag, select a chip, record its color, and then return the chip to the bag. Repeat the process 4 more times to obtain a simulated outcome. Then repeat this process many times and find the proportion of the simulations in which exactly 4 blues are selected to estimate the probability.	*B*

- *How do you think the experimental probabilities that would be collected from these simulations will compare to their theoretical probabilities?*

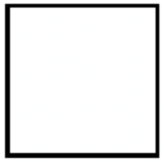
To extend students' thinking, have students choose one or more of the simulations and conduct the experiment and record their results and the experimental probability. How does it compare to the expected theoretical probability?

15.6.5 Wrap-Up: Reflection

Component Overview: *Students self-reflect on simulating the probability of events.*

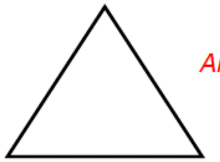
15.6.5 Wrap-Up: Reflection

1. What is something that is now "squared away" in your mind about simulating the probability of events?



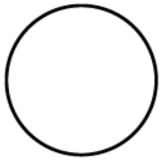
Answers will vary.

2. What are three "points" that are important about simulating the probability of events?



Answers will vary.

3. What is something that is still "circling" your brain about simulating the probability of events?



Answers will vary.

In the next lesson, students will continue to use simulations to find the probability of events. Look for common themes in what students are struggling with to address in the next lesson.

15.7 Experimental and Theoretical Probabilities from Simulations

Lesson Introduction

Benchmarks	MA.7.MA.DP.2.3 Find the theoretical probability of an event related to a simple experiment. MA.7.MA.DP.2.4 Use a simulation of a simple experiment to find experimental probabilities and compare them to theoretical probabilities.	Learning Target	• I can determine the probability of an event from a simulation and compare it to the theoretical probability of the event.

	Building On	Working Toward
Benchmark Coherence	N/A	MA.8.DP.2.2 Find the theoretical probability of an event related to a repeated experiment. MA.8.DP.2.3 Solve real-world problems involving probabilities related to single or repeated experiments, including making predictions based on theoretical probability.

Essential Questions	• How do you determine a simple experiment that could be used to simulate the probability of an event? • How does the probability from a simulation compare to the theoretical probability of the event?
Lesson Narrative	In this lesson, students expand their knowledge of theoretical and experimental probabilities through simulations. Students practice conducting simulations to determine experimental probabilities. To start the lesson, students determine different simple experiments that could be used to simulate a probability. Students then conduct those experiments and record the data. They then use the data and compare the theoretical and experimental probabilities.

Component Summary	15.7.1 Warm-Up (~5 min)	Visualize probabilities (SE pg. 432)	15.7.3 Your Turn! (15 min)	Practice finding experimental and theoretical probabilities (SE pg. 434)
	15.7.2 Collaboration (20 min)	Practice simulating probabilities (SE pg. 432)	15.7.4 Wrap-Up (~5 min)	Self-reflect on the learning targets from the unit (SE pg. 435)

	Glossary	Materials	Additional Preparation
Resources	Key Terms: N/A Additional Terms: • Experimental probability • Theoretical probability • Simple experiment	• 6-sided die • 12 equal-sized paper slips numbered 1-12 • Standard deck of 52 cards • 4-section spinners	• If any materials are not available, electronic die or simulators can be used instead.

<table>
<tr><td rowspan="4">Teacher Tips</td><th>Common Misconceptions</th><th>Things to Consider</th></tr>
<tr><td>In 15.7.3 Your Turn!, students may only count the cards between 11 and 25, not including those numbers. Consider prompting students to write out the numbers and count how many cards there would be.
Students may confuse theoretical and experimental probability.</td><td>Consider providing equations for solving for theoretical probability and posting them on the board for students to reference.</td></tr>
<tr><th>Literacy and Language Routines</th><th>Mathematical Thinking and Reasoning (MTR) Standard</th></tr>
<tr><td>Think-Pair-Share
Consider implementing a Think-Pair-Share for question 2 in 15.7.2 Collaboration. Students can work with their partner to come up with the different simple experiments that could be used to simulate Jimmy’s three-point shot probability.
As a class, partners can share what they came up with and discuss how each probability should be equivalent to $\frac{1}{4}$.</td><td>MA.K12.MTR.7.1: Real-World Contexts
Students may have experience with seeing probabilities used in the real world, for example, in sports. Use 15.7.2 Collaboration and conversation about statistics in sports to bridge probability and statistics, which are used in a variety of industries in the “real world.”</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">In 15.7.2 Collaboration, consider scaffolding by reminding students to goal is to create a fraction equivalent to $\frac{2}{3}$. You can provide students with the following fill in the blank equations to help…
For a six-sided die, $\frac{2}{3} = \frac{}{6}$
For a standard 52-card deck, $\frac{2}{3} = \frac{}{52}$</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
</table>

15.7.1 Warm-Up: Visualizing Probabilities

Component Overview: *Students visualize probabilities using double number line diagrams before interpreting a probability verbally.*

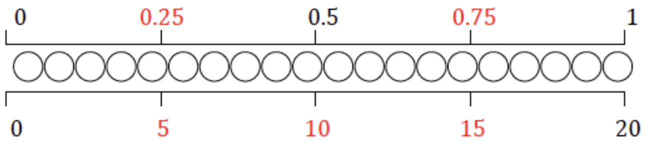
15.7.1 Warm-Up: Visualizing Probabilities

Ashley places 2 red marbles, 3 blue marbles, 4 yellow marbles, 5 orange marbles, and 6 purple marbles in a bag.

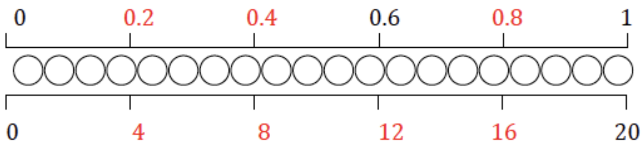
A representation of the marbles is shown.

1. Complete the double number line diagrams below by writing values at each tick mark. The top line represents probabilities, and the bottom line represents the number of marbles in the bag.

a.



b.



2. Write a color on each blank to complete the statements.

The probability of pulling a ______*yellow*______marble out of the bag without looking is 0.2. The probability of pulling an _____*orange*_____ marble out of the bag without looking is 0.25.

Prompt students to notice into how many sections the double number line is broken to determine the missing pieces.
If one whole is broken into 4 equal pieces, what is each piece worth?

15.7.2 Collaboration: Simulating Probabilities

__Component Overview:__ Students practice simulating probabilities by coming up with simulations using different simple experiments that could be used to describe Jimmy Butler's probability of making a three-point shot.

15.7.2 Collaboration: Simulating Probabilities

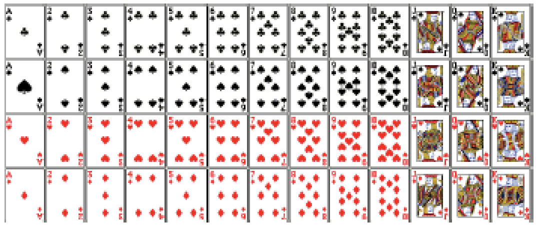
1. In the 2019-2020 NBA regular season, Miami Heat player Jimmy Butler made $\frac{1}{4}$ of the three-point shots he attempted. What was the probability of Jimmy Butler making a three-point shot in the 2019-2020 regular season, represented as a decimal?
   0.25

2. Work with your partner to complete each statement to describe ways Jimmy's three-point shot probability could be simulated using different simple experiments.

   a. Create a spinner with 4 equal section(s), where 1 section(s) represent(s) Jimmy Butler making a three-point shot and the other(s) represent(s) misses.

   b. Place twelve equal-sized slips of paper with the numbers $1-12$ written on them in a bag. Papers with the numbers *answers will vary (any three numbers)* represent Jimmy Butler making a three-point shot and the others represent misses.

   c. Use a standard deck of 52 cards where *Answers will vary. (Any 13 cards)* represent Jimmy Butler making a three-point shot and the other cards represent misses.



Think-Pair-Share
Consider implementing a Think-Pair-Share for question 2. Students can work with their partner to come up with the different simple experiments that could be used to simulate Jimmy's three-point shot probability.
As a class, partners can share out what they came up with and discuss how each fraction should be equivalent to $\frac{1}{4}$.

Prompt students to use the visual of the cards to help.

15.7.2 Collaboration: Simulating Probabilities (continued)

3. Suppose in the first post-season game that year, Jimmy Butler attempted 5 three-point shots. Choose one of the simulations from the previous question and work with your partner or group to simulate the number of three-point shots Jimmy made.
   Answers will vary.
   a. Repeat the simulation 5 times and record your results in the table.

	Shot 1	Shot 2	Shot 3	Shot 4	Shot 5	Fraction Made
Simulation 1						$\frac{\quad}{5}$
Simulation 2						$\frac{\quad}{5}$
Simulation 3						$\frac{\quad}{5}$
Simulation 4						$\frac{\quad}{5}$
Simulation 5						$\frac{\quad}{5}$

   b. Based on your group's simulations, estimate the probability that Jimmy Butler made $\frac{2}{5}$ of the three-point shots he attempted in the game.
   Answers will vary.

   c. Jimmy Butler actually made 3 out of the 5 three-point shots he attempted in the game. Explain how that compares to your group's simulation results.
   Answers will vary.

Students will need to have spinners, papers, and standard decks of cards that they can use to perform the simulation.

If supplies are not available, a random number generator or virtual simulator can be used online.

Why is it that the experimental probabilities are not always the same as each other or to the theoretical probability?

4. In the 2019-2020 NBA regular season, Orlando Magic player Aaron Gordon made $\frac{2}{3}$ of free-throw shots attempted. Work with your partner to describe how each of the following could be used to simulate his attempted free-throw shots.

   a. A fair 6-sided die
   Answers will vary. Any four numbers on a six-sided dice will accurately simulate the number of free throws that Aaron Gordon will make.

   b. A standard 52-card deck
   Answers will vary. The number 52 is not divisible by three, so one card will need to be taken out of the deck. If one card is taken out, then any 34 cards in the deck would simulate the number of free throws that Aaron Gordon will make.

   c. Different colored marbles placed in a bag
   Answers will vary. Two-thirds of the colored marbles chosen would simulate the number of free throws that Aaron Gordon will make.

Consider scaffolding by reminding students the goal is to create simulations with probabilities equivalent to $\frac{2}{3}$. You can provide students with the following fill in the blank equations to help…

For a six-sided die, $\frac{2}{3} = \frac{}{6}$.

For a standard 52-card deck, $\frac{2}{3} = \frac{}{52}$.

15.7.3 Your Turn!

Component Overview: *Students continue conducting simulations to determine experimental probabilities. This time, students use a 6-sided die and record the data and use it to find different probabilities.*

15.7.3 Your Turn!

1. Suppose a fair six-sided die numbered 1 through 6 is rolled 12 times.

   a. Of those 12 rolls, how many do you expect will land on the number 3?

   2 times out of 12 rolls

   b. Roll a standard six-sided die 12 times and record the number on the top of the cube when it lands.

Roll 1	Roll 2	Roll 3	Roll 4	Roll 5	Roll 6	Roll 7	Roll 8	Roll 9	Roll 10	Roll 11	Roll 12

   c. What is the theoretical probability of rolling a 3?

   $\frac{1}{6}$

   d. Describe how the theoretical probability compares to your experimental probability.

   Answers will vary.

   e. Review your results and your partner's results. Explain why the results are not the same.

   Answers will vary. The more times that the dice is rolled, the closer the experimental probability will get to the theoretical.

Students will either need a die or access to a digital die.

Consider a reminder of how to find theoretical probability.

$$\frac{\textit{number of desired outcomes}}{\textit{total number of possible outcomes}}$$

Consider posting this formula on the board for students to reference.

Does the number of trials affect how close the theoretical and experimental probabilities are?

15.7.3 Your Turn! (continued)

2. Consider the events of rolling a fair 10-sided die and getting an odd number or flipping a coin and having it land tails-up. Explain which event is more likely.
The events are equally likely because the theoretical probability of tossing a coin and it landing tails-up is $\frac{1}{2}$. The theoretical probability of rolling a ten-sided dice and landing on an odd number is $\frac{5}{10}$ that reduces to $\frac{1}{2}$.

3. Noelle picks a card at random from a stack that has cards numbered 11 through 25.

   a. Determine the sample space.
   11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25
   The sample space is fifteen cards.

   b. Find each probability.

   $P(\text{even}) = \frac{7}{15}$ *or* $\approx 47\%$

   $P(\text{multiple of } 5) = \frac{3}{15}$ *or* 20%

   $P(\text{two digit number}) = \frac{15}{15}$ *or* 100%

   $P(\text{number less than } 10) = \frac{0}{15}$ *or* 0%

   c. Noelle used a simulator to run the experiment 300 times. She selected an odd-numbered card 154 times. Compare the experimental probability to the theoretical probability of selecting an odd-numbered card.
   The experimental probability is $\frac{154}{300}$ *or* 51.3%*. The theoretical probability of picking an odd number is* $\frac{8}{15}$ *or* 53.3%*.*

Students may only count the cards between 11 and 25, not including those numbers. Consider prompting students to write out the numbers and count how many cards there would be.

To extend students' thinking, have them research a sports player's statistics of their choosing.
Students can then come up with a simple experiment that could be used to simulate the probability researched.

15.7.4 Wrap-Up: Reflection

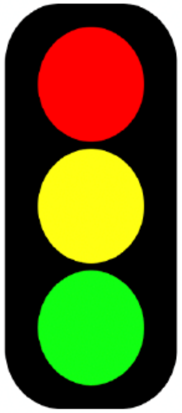
Component Overview: *Students reflect on their understanding of concepts addressed throughout the unit.*

15.7.4 Wrap-Up: Reflection

In this unit, you have learned multiple concepts about probability, including those listed below.

- Determine the sample space for a simple experiment
- Express probabilities as fractions, decimals, and percentages
- Interpret and compare the probability of chance events
- Find the theoretical probability of an event related to a simple experiment
- Compare experimental probability from simulations to theoretical probability

1. Use the stoplight reflection below to share how you are feeling about each of these concepts at this point.



I still feel stuck on how to …
Answers will vary.

I'm starting to get the hang of …
Answers will vary.

I feel confident and ready to go with …
Answers will vary.

For an alternative approach to the Wrap-Up, have students use red, yellow, and green highlighters to highlight the concepts listed based on how they feel about each one.